

```
In [24]: import pandas as pd

# Load dataset
df = pd.read_csv("train.csv")

# Check first few rows
df.head()

# Basic structure
df.info()

# Statistical summary
df.describe()

# Check missing values
df.isnull().sum()

# Value counts for categorical columns
df["sex"].value_counts()
df["embarked"].value_counts()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
# Column Non-Null Count Dtype
---
0 PassengerId 891 non-null int64
1 Survived 891 non-null int64
2 Pclass 891 non-null int64
3 Name 891 non-null object
4 Sex 891 non-null object
5 Age 714 non-null float64
6 SibSp 891 non-null int64
7 Parch 891 non-null int64
8 Ticket 891 non-null object
9 Fare 891 non-null float64
10 Cabin 204 non-null object
11 Embarked 891 non-null object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.74 KB

Out[24]: Embarked
C 644
S 168
Q 77
Name: count, dtype: int64

In [22]: df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
# Column Non-Null Count Dtype
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```

Observation: Missing Data & Data Types

- The dataset contains 891 rows and 12 columns.
- Some columns have missing values:
 - Age has 177 missing entries.
 - Cabin has many missing values (687 out of 891).
 - Embarked has 2 missing entries.
- Columns like Name, Sex, Ticket, Cabin, and Embarked are of type object, meaning they are strings (categorical data).
- Columns like Age, Fare are of type float.
- Survived, Pclass, SibSp, and Parch are of type int, indicating discrete numeric values.

```
In [18]: df.isnull().sum()

Out[18]: PassengerId    0
Survived              0
Pclass               0
Name                 0
Sex                  0
Age                 177
SibSp                0
Parch               0
Ticket              0
Fare                 0
Cabin               687
Embarked             2
dtype: int64
```

Missing Value Summary

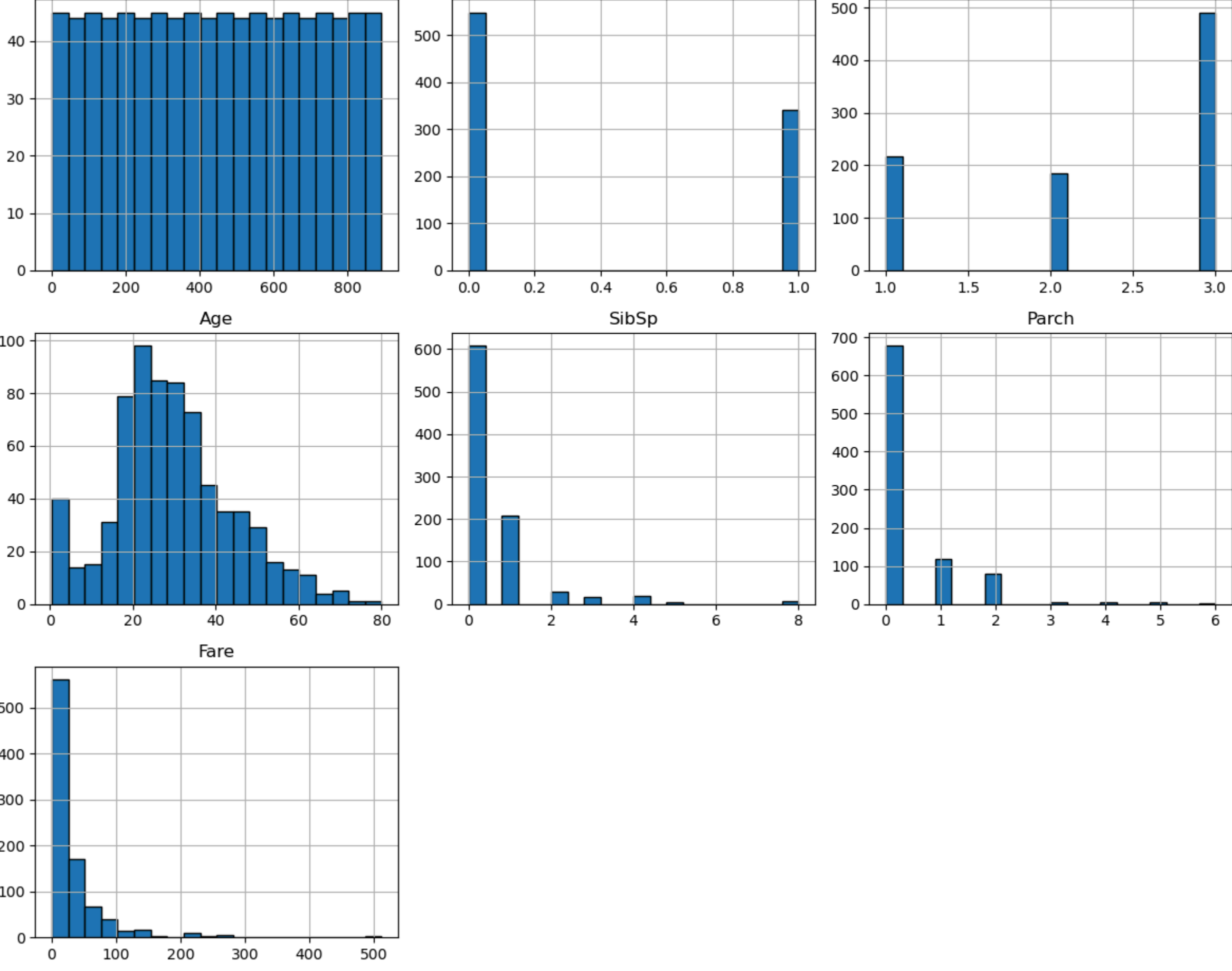
- The Cabin column has the highest number of missing values (687), which is more than 75% of the dataset. This might be dropped or filled with a new category like "Unknown".
- Age is an important column and has 177 missing values. We'll need to impute (fill) these values — maybe with median or using group-based imputation.
- Only 2 values are missing in Embarked, which can easily be filled using the mode (most frequent value).

VISUAL EXPLORATION : Univariate Analysis – Histograms for Numeric Columns

```
In [38]: import matplotlib.pyplot as plt

df.hist(figsize=(12, 10), bins=20, edgecolor='black')
plt.suptitle("Histograms of Numeric Columns", fontsize=10)
plt.tight_layout()
plt.show()
```

Histograms of Numeric Columns



Histogram Observation

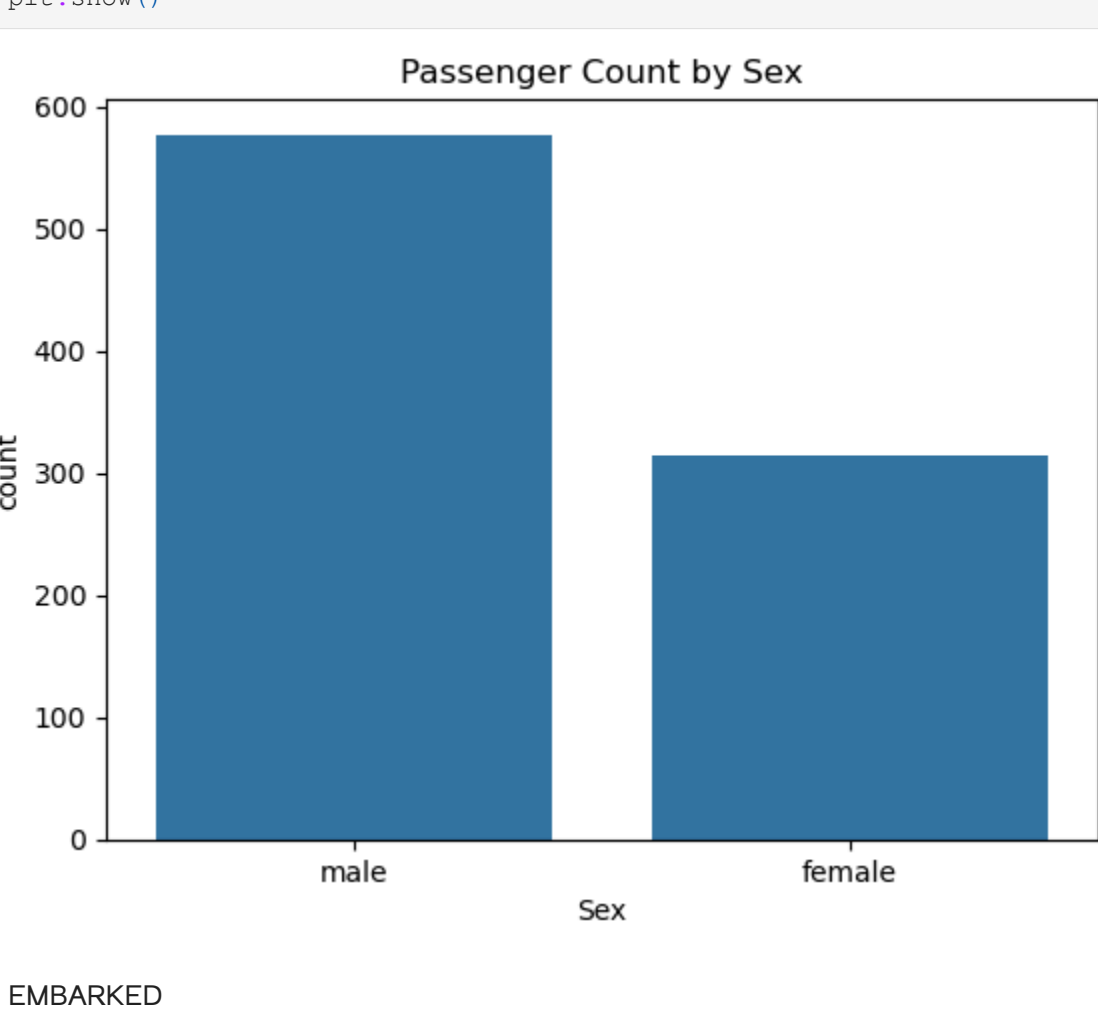
- Age is right-skewed, with many passengers in their 20s and 30s.
- Fare has a long tail with a few passengers paying very high prices.
- SibSp and Parch are concentrated around 0, meaning most people traveled alone or with one family member.

Count Plots for Categorical Columns

SEX

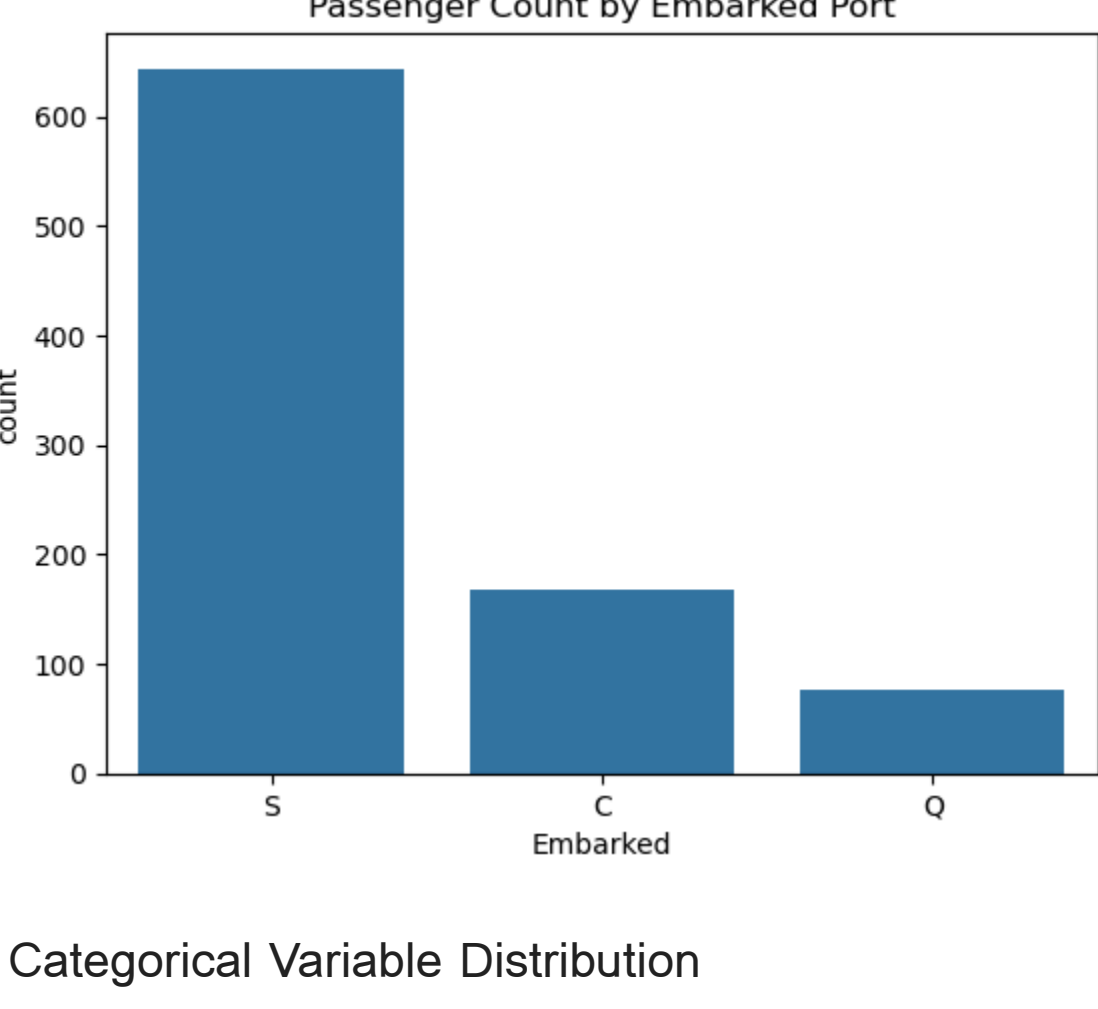
```
In [41]: import seaborn as sns

sns.countplot(data=df, x="Sex")
plt.title("Passenger Count by Sex")
plt.show()
```



EMBARKED

```
In [44]: sns.countplot(data=df, x="Embarked")
plt.title("Passenger Count by Embarked Port")
plt.show()
```



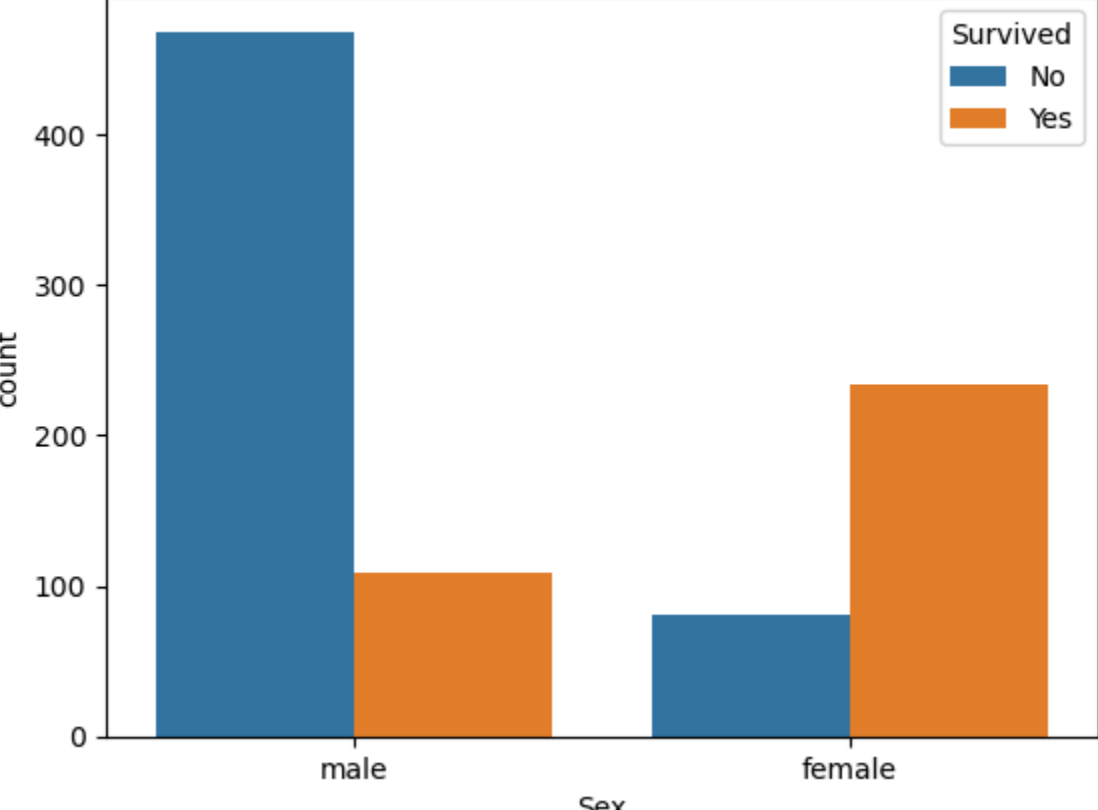
Categorical Variable Distribution

- More males (Sex) than females in the dataset.
- Most passengers embarked from port S (Southampton), followed by C (Cherbourg).

BIVARIATE ANALYSIS

Survival by Gender

```
In [52]: sns.countplot(data=df, x="Sex", hue="Survived")
plt.title("Survival Count by Gender")
plt.legend(title="Survived", labels=["No", "Yes"])
plt.show()
```

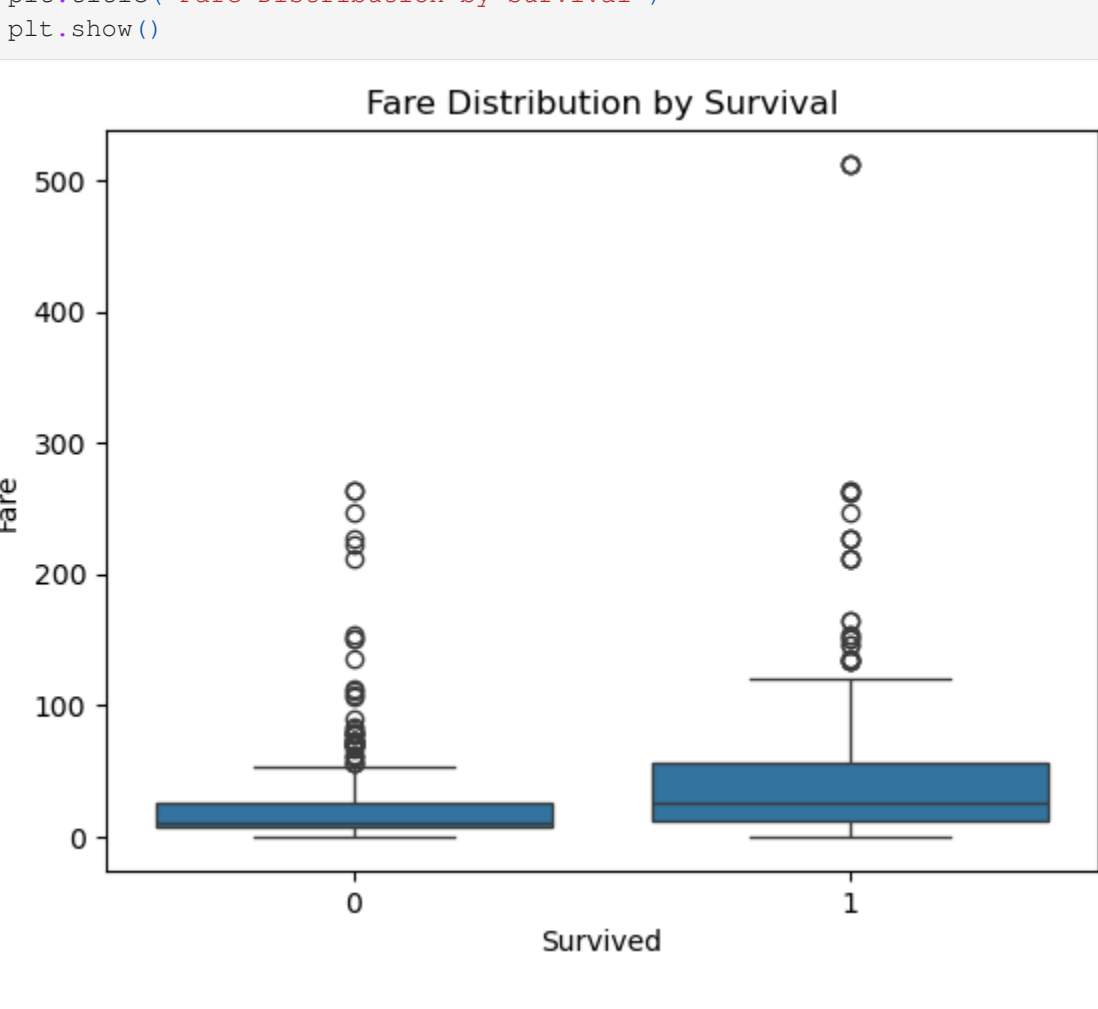


Survival by Gender

- Females had a significantly higher survival rate than males.
- Most of the male passengers did not survive.

Boxplot: Fare vs Survived

```
In [56]: sns.boxplot(data=df, x="Survived", y="Fare")
plt.title("Fare Distribution by Survival")
plt.show()
```



Fare vs Survival

- Passengers who paid higher fares were more likely to survive.
- Non-survivors mostly had lower fare tickets.

Violin Plot: Age vs Survived

```
In [62]: sns.violinplot(data=df, x="Survived", y="Age")
plt.title("Age Distribution by Survival")
plt.show()
```



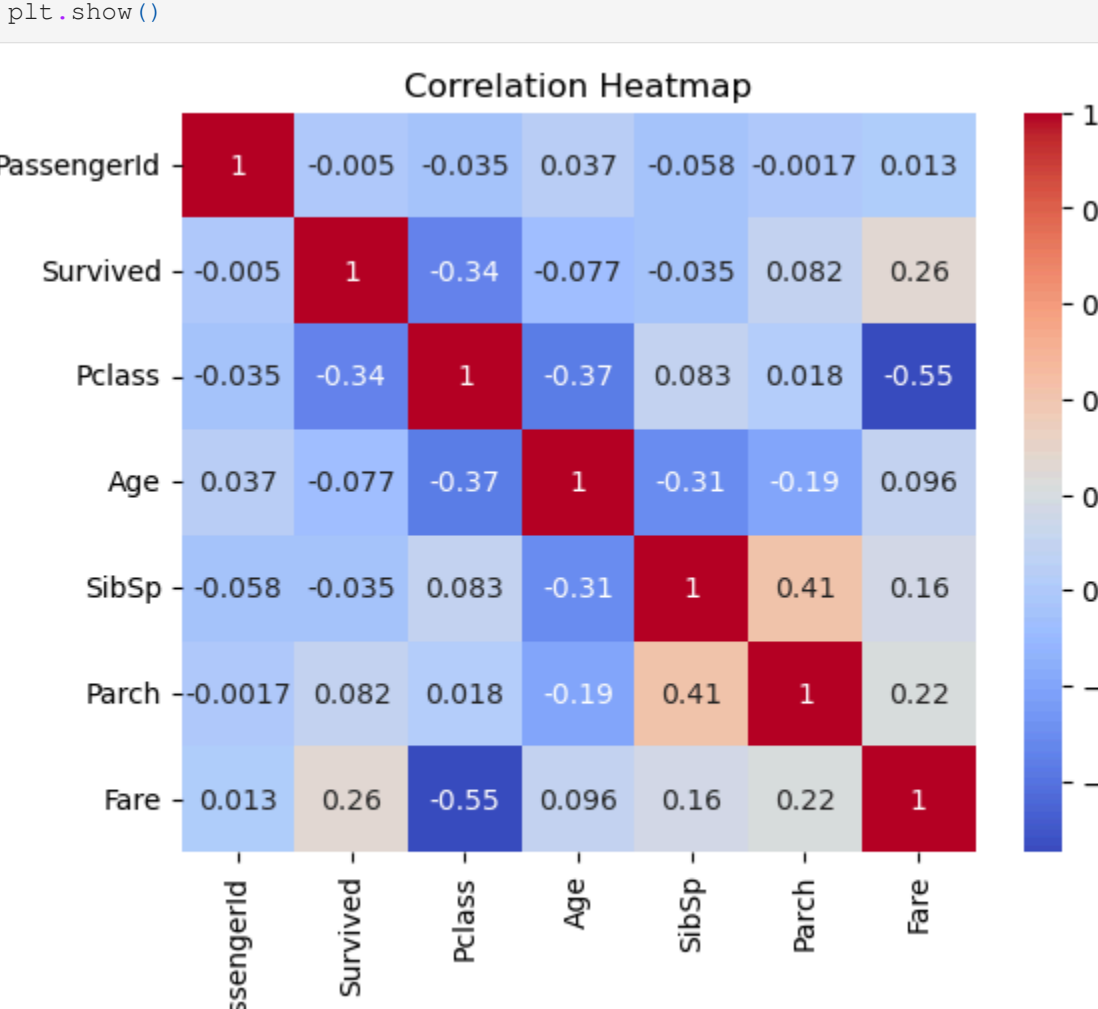
Age vs Survival

- Children and younger adults were more likely to survive.
- Older passengers have lower survival density.

CORRELATION & MULTIVARIATE ANALYSIS

Heatmap of Correlation

```
In [67]: corr = df.corr(numeric_only=True)
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```

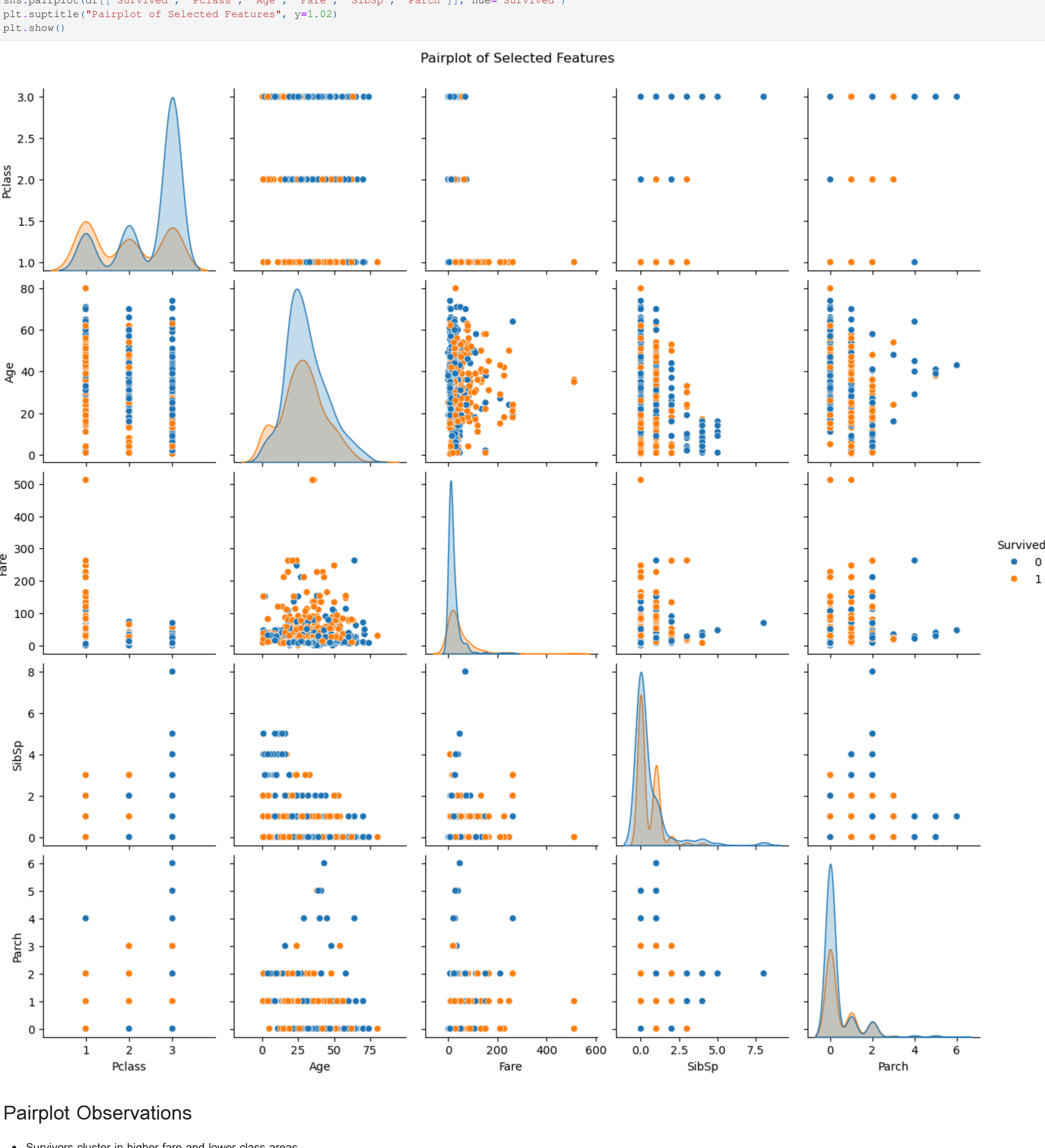


Correlation Heatmap

- Fare and Pclass show some correlation with Survived.
- SibSp and Parch are also slightly correlated.
- No extreme multicollinearity observed.

Pairplot for Visual Relationships

```
In [72]: sns.pairplot(df[['Survived', 'Pclass', 'Age', 'Fare', 'SibSp', 'Parch']], hue='Survived')
plt.suptitle("Pairplot of Selected Features", y=0.2)
plt.show()
```



Pairplot Observations

- Survivors cluster in higher fare and lower class areas.
- Most passengers fall into low SibSp and Parch groups.
- There is visible separation between survivors and non-survivors in some features.

Handle All Missing Data

```
In [85]: # Fill missing Age with median (safe version)
df["Age"] = df["Age"].fillna(df["Age"].median())

# Fill missing Embarked with mode (safe version)
df["embarked"] = df["embarked"].fillna(df["embarked"].mode()[0])

# Drop Cabin column only if it exists (to avoid KeyError)
if "Cabin" in df.columns:
    df = df.drop(columns="Cabin")

# Final check to confirm all missing values are handled
print(✓ Missing values after cleanup: 0)
print(df.isnull().sum())

Missing values after cleanup:
PassengerId    0
Survived        0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtype: int64
```

Data Cleaning Summary

- Filled missing Age values with the median.
- Filled missing Embarked values with the mode (most common port).
- Dropped Cabin column only if present to avoid errors.
- Dataset is now free of missing values and ready for analysis or modeling.

Summary of Findings:

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- Passenger Gender: Female passengers had a much higher survival rate than males.
- Passenger Class: Passengers in 1st class had significantly better survival chances than those in 2nd or 3rd class.
- Fare Paid: Higher fare correlated with higher survival — suggesting wealthier passengers had better access to lifeboats or priority.
- Age: Children and younger adults had slightly better survival chances.
- Family Size: Most passengers traveled alone or with one relative, this may have a mild effect on survival.
- Embarked Port: Most passengers boarded at Southampton (S), but embarkation port had a relatively small influence on survival.

Data Cleanup Summary:

- Missing values in Age and Embarked were imputed.
- The Cabin column was dropped due to excessive missing data.

